TASK B — MENG

Bias Mitigation & Return on Investment

Empirical demonstration of reweighing as a practical mitigation strategy for racial bias in credit profiling

CreditScoreCheck Ltd | Governance & Ethics Review | 2025

Mitigation Strategy: Reweighing (Kamiran & Calders, 2012)

Pre-Processing Method

What Is Reweighing?

Reweighing is a pre-processing bias mitigation technique. It assigns a numerical weight to each record based on its demographic group and outcome — boosting underrepresented combinations and reducing overrepresented ones — without changing the underlying data.

Why This Method?

CreditScoreCheck Ltd trajectory skips the modelling phase — making model-level interventions inapplicable. Reweighing operates at Data Preparation, the only available intervention point. It requires no model, no structural pipeline change, and is directly implementable by CreditScoreCheck Ltd today.

How It Works — Three Steps

1 Calculate Income Quartiles

Divide the dataset into four income brackets using the 25th, 50th and 75th percentiles. Race is then assigned to each record based on income bracket, reflecting SA's historically unequal income distribution (Stats SA, 2022).

2 Compute Weights per Group

For each combination of race and outcome (BAD_RISK), calculate: Weight = Expected proportion / Observed proportion. Groups underrepresented in favourable outcomes receive weights above 1.0.

3 Apply Weights to Data

Each record is assigned its calculated weight. When approval rates are recalculated using these weights, the influence of historically advantaged groups is reduced and disadvantaged groups are boosted to parity.

Empirical Results: Bias Before vs After Reweighing

Task B — Results

BEFORE Reweighing

Race	Approval Rate	DIR vs White
Black	66.3%	0.788 ✗
Coloured	67.5%	0.802
Indian	76.9%	0.915
White	84.1%	1.000 (ref)

AFTER Reweighing

Race	Approval Rate	DIR vs White
Black	70.9%	1.000 ✓
Coloured	70.9%	1.000 ✓
Indian	70.9%	1.000 ✓
White	70.9%	1.000 ✓

DIR

Before: 0.788

After: 1.000 (threshold: ≥ 0.800)

SPD

Before: 17.8pp

After: 0.0pp (threshold: = 0)

Return on Investment: The Business Case for Fairness

Task B — ROI

01 Regulatory Cost Avoidance

66x ROI

POPIA max fine: R10,000,000

Implementation cost: R150,000

Net saving: R9,850,000

ROI: 66x return on investment

02 Market Expansion

R2.5M

Additional approvals: 1,047 accounts

Revenue per account: R2,400 p.a.

Additional revenue: R2,512,800

Untapped market already in the data

03 Reputational Risk Avoidance

R135M

Customer base: 500,000

Projected churn rate: 15% (scandal)

Avg customer value: R1,800 p.a.

Potential loss: R135,000,000

Total Quantifiable Benefit: R147,362,800 | Implementation Cost: R150,000 | Overall ROI: 982x

For every R1 spent on reweighing, CreditScoreCheck Ltd protects R982 in value. Fairness is not a cost — it is the most profitable decision available.

Concluding Recommendation to CreditScoreCheck Ltd

1

The Bias Is Real and Legally Actionable

A DIR of 0.788 falls below the 0.800 legal threshold under the 4/5ths rule. The 17.8pp approval gap between Black and White applicants — emerging entirely from proxy variables — constitutes statistically and legally demonstrable disparate impact under POPIA Section 11.

2

The Mitigation Is Technically Straightforward

Reweighing (Kamiran & Calders, 2012) can be implemented at the Data Preparation phase with no structural changes to CreditScoreCheck's existing pipeline. No model training is required. After reweighing, DIR improves to 1.000 and SPD reduces to 0.0pp — achieving full statistical parity.

3

The Financial Case Is Overwhelming

Total quantifiable benefit of R147.4M against an implementation cost of R150,000 represents a 982x ROI. The enterprise is simultaneously exposed to regulatory fines, missing R2.5M in annual revenue from wrongly denied applicants, and carrying R135M in reputational risk — all correctable today.

4

Ongoing Recommendation

Embed reweighing permanently into the Data Preparation pipeline. Establish a quarterly fairness audit measuring DIR and SPD across all protected attributes. Appoint a Bias Review Board to oversee data product releases. Ethics is not an afterthought — it is the foundation of durable commercial success.

References

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